Andres Tiko

Machine learning practitioner, Software developer, tech lead & CTO, retired particle physicist

e-mail | Linkedin

Skills

- Software development and architecture (mainly with Python-based stacks)
- Data analysis and machine learning
- Research and problem solving
- Technical leadership

Employment

`2024 Jun - present` **Staff Machine Learning Engineer, Zendesk**

- Developing, maintaining and monitoring ML pipelines for high-volume data processing (Apache Beam, Celery, Redis, BigQuery, Kubernetes, Datadog, Grafana, GCP)
- Developed a new service that enables customers to create their own prompt-based categories to evaluate customer service tickets (Langchain, LLMs, FastAPI)

`2022 - 2024` **Technical Lead -> CTO & co-founder, Soulie**

- Leading the development of an app helping users spend their time mindfully by recommending content based on their goals and preferences and encouraging focusing instead of mindless scrolling
- Leading the development from prototype to public release
- Managing the technical team
- Architecting and building the backend systems and APIs (Python, FastAPI, MongoDB, Docker, AWS)
- Building the infrastructure in AWS etc.
- Improving the machine learning systems for making recommendations
- Started as a tech lead, joined the co-founders team as CTO

`2022 Jan - Oct` **AI & Backend Developer, Zurich Insurance Company Ltd**

- Continuing the work developing virtual assistants after AlphaBlues was acquired by Zurich Insurance Company (as described below)

`2019 - 2021` **Back-end developer, data scientist, AlphaBlues, a company developing automated customer service solutions using virtual assistants**

- Introduced modern deep learning based models for natural language processing (NLP) for detecting the intent of a user interacting with a virtual assistant.
- Architected and developed the pipelines and back-end systems for training and deployment of those models.
- Created tools to help users add training data where it is needed as well as gain insights about how the system is performing.
- Developing asynchronous web services and improving their security

- Main tools: python, fast.ai, Pandas, PyTorch, Tornado web server, MongoDB, Docker, AWS

2021 **Data scraping project for University of Tartu research project "Academic Writing in the Baltic States: Rhetorical Structures through culture(s) and languages"**

- Wrote the necessary scripts and scraped various academic writing content (tens of thousands of files) for the research project

2017 - 2019 **INFN Fellowship, for non-Italian citizens for research activity in Experimental Physics, Italian National Institute for Nuclear Physics, Padova**

- Became the lead author for the analysis "Search for nonresonant Higgs boson pair production in the bbbb final state" and lead it to publication: Journal of High Energy Physics 04 (2019) 112 - the analysis is based on novel idea for modelling the hard to simulate QCD multijet process using clustering and uses classification methods for signal-background separation.
- Took care of the operation of the CMS detector as a data acquisition (DAQ) shifter
- Main tools used: python, numpy, scikit-learn, Pandas, XGBoost, C++

2010 - 2017 **Engineer and Researcher, KBFI (National Institute for Chemical Physics and Biophysics), Tallinn, Estonia**

- Worked on the analysis of data collected by the CMS experiment, resulting in multiple publications and the completion of my PhD thesis

2009 - 2010 **ACE Software Development intern with AIESEC, Tata Consultancy Services, Mumbai, India**

- Developed enterprise resource planning software for manufacturers (Java, SQL).

2006 - 2007 **Developer of Software and Services, Mobi Solutions**

- Worked on full-stack development of a platform to create SMS-based services, implementing new features and improving existing functionality (Java, SQL, Struts).
- Developed the backend for a new version of the platform - Fortumo (Java, SQL).

Education

2010 - 2016 **PhD in Physics University of Tartu and National Institute of Chemical Physics and Biophysics**

- Thesis: "Measurement of Top Quark Properties in the CMS experiment at the LHC", supervisors Andrea Giammanco, Mario Kadastik, Martti Raidal

2007 - 2009 **MSc in Computer Science, University of Tartu**

- Thesis: "Ripple-Down Rule Sets and MDL Learning", supervisors Jaak Vilo and Meelis Kull - used the minimal description length (MDL) principle to learn rule sets with exceptions that are also naturally understandable to humans.

2003 - 2007 **BSc in Physics, University of Tartu**

- Thesis: "Superconductive Properties in a Two-gap Model with a Metallic Spectrum", supervisor N. Kristoffel